



NAME:
HABIB ID:
SECTION:

Max Marks: 10

Time: 10 minutes

LINEAR ALGEBRA

SPRING 2026

QUIZ 5

NOTE: No devices, calculators, or phones are allowed while you take the quiz.

1. If A is an $n \times n$ matrix and E is an $n \times n$ elementary matrix, then

$$\det(EA) = \det(E) \det(A).$$

Given that E results from interchanging two rows of I_n .

[6]

2. Let

$$A = \begin{pmatrix} b+c & c+a & b+a+2c \\ a & b & a+b \\ 1 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ without using cofactor expansion.

[4]



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1. If B is an $n \times n$ matrix and E is an $n \times n$ elementary matrix, then

$$\det(EB) = \det(E) \det(B).$$

Given that E results from adding a multiple of one row of I_n to another.

[6]

2. Let

$$B = \begin{pmatrix} b+a+2c & c+a & b+c \\ a+b & b & a \\ 2 & 1 & 1 \end{pmatrix}$$

Find $\det(B)$ without using cofactor expansion.

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1. If C is an $n \times n$ matrix and E is an $n \times n$ elementary matrix, then

$$\det(EC) = \det(E) \det(C).$$

Given that E results from multiplying a row of I_n by a scalar k .

[6]

2. Let

$$C = \begin{pmatrix} b+a+2c & b+c & c+a \\ a+b & a & b \\ 2 & 1 & 1 \end{pmatrix}$$

Find $\det(C)$ without using cofactor expansion.

[4]



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1. EB is exactly the matrix you get by applying the same row operation to B :

$$EB = "B \text{ with } R_i \text{ replaced by } R_i + kR_j".$$

Adding a multiple of one row to a different row does not change the determinant.

So,

$$\det(EB) = \det(B).$$

is obtained from I_n by $R_i \leftarrow R_i + kR_j$, and this operation does not change determinants,

$$\det(E) = \det(I_n) = 1.$$

Using the two results:

$$\det(EB) = \det(B) = 1 \cdot \det(B) = \det(E)\det(B).$$

if EB is obtained from B by swapping two rows, then

$$\det(EB) = -\det(B).$$

Now look at E itself. Since E is I_n with two rows interchanged, it is the identity matrix with a row swap applied. Swapping two rows of I_n also flips the sign of its determinant:

$$\det(E) = \det(\text{row-swapped } I_n) = -\det(I_n) = -1,$$

- $\det(EB) = -\det(B)$
- $\det(E) = -1$

Therefore,

$$\det(EB) = -\det(B) = (-1)\det(B) = \det(E)\det(B).$$

2.
$$A = \begin{pmatrix} b+c & c+a & b+a+2c \\ a & b & a+b \\ 1 & 1 & 2 \end{pmatrix}$$

Let the columns be C_1, C_2, C_3 . Then:

$$C_3 = C_1 + C_2$$

because



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$$C_1 + C_2 = \begin{pmatrix} (b+c) + (c+a) \\ a+b \\ 1+1 \end{pmatrix} = \begin{pmatrix} b+a+2c \\ a+b \\ 2 \end{pmatrix} = C_3.$$

So the columns are linearly dependent, hence

$$\det(A) = 0.$$